AI Search Agent using n8n

1. Introduction

Artificial Intelligence agents are systems that can automatically perform tasks by interacting with users, accessing external tools, and processing information intelligently. This project implements an **AI-powered search assistant** using an automation platform that integrates a language model with web search capabilities.

The system is designed to receive a user query through a chat interface, process the request using an AI model, and fetch relevant information from the internet. The AI agent can also remember previous interactions using a memory component, which improves contextual understanding during conversations.

The workflow is developed using **n8n**, an open-source workflow automation tool that allows the integration of APIs, AI models, and external services.

2. Objectives

The main objectives of this project are:

- To design an **AI-based conversational search system**.
- To integrate a **large language model** for natural language understanding.
- To enable **real-time web search capabilities** using a search API.
- To store and manage conversation context using memory.
- To automate the process using a **workflow automation platform**.

3. System Architecture

The system architecture consists of multiple interconnected components that process the user query step by step.

Components

1. Chat Trigger Node
2. AI Agent Node
3. Chat Model (LLM)
4. Memory Module
5. Web Search Tool

The workflow begins when a user sends a message. The AI agent processes the request using the language model and retrieves additional information from the web search API when required.

4. Tools and Technologies Used

Workflow Automation

- n8n
  Used to create automated workflows and integrate AI services.

AI Model Provider

- Groq
  Provides high-speed inference for large language models.

Language Model

Example models used:

- llama-3.1-8b-instant
- llama-3.3-70b-versatile

These models are responsible for generating intelligent responses.

Web Search API

- SerpApi

Used to retrieve real-time search results from search engines.

Programming Environment

- Browser-based visual workflow editor
- API-based integrations

5. Workflow Design

The workflow consists of several nodes connected together to perform the AI search process.

Step 1: Chat Message Trigger

The workflow begins with the **When Chat Message Received** node. This node listens for user input from the chat interface.

Function:

- Captures user query
- Initiates workflow execution

Step 2: AI Agent

The AI Agent acts as the **central controller** of the system.

Responsibilities:

- Understand user queries
- Decide whether a tool is required
- Send requests to the language model
- Integrate search results into the response

The AI agent combines the power of language models with external tools.

Step 3: Chat Model

The chat model processes natural language queries and generates responses.

Model used:

- llama-3.1-8b-instant

Functions:

- Natural language understanding
- Response generation
- Tool usage decision making

Step 4: Memory Module

The **Simple Memory** node stores conversation context.

Purpose:

- Maintain chat history
- Provide contextual understanding
- Improve response relevance

Example:
User: "Suggest tourist places in Kollam"
Next question: "Which one is best?"

The memory helps the agent understand that the second question refers to the previously suggested places.

Step 5: Search Tool Integration

The search tool is connected to the AI agent using SerpAPI.

Functions:

- Fetch real-time web data
- Retrieve search results
- Provide updated information

Example query:
"Best restaurants in Hyderabad"

The agent uses the search tool to gather relevant results and summarize them.

6. Workflow Execution Process

The system executes the following sequence:

1. User sends a message through the chat interface.
2. The chat trigger node activates the workflow.
3. The AI agent receives the query.
4. The agent sends the request to the language model.
5. If external data is needed, the agent calls the search API.
6. The search results are processed.
7. The AI model generates a final response.
8. The response is displayed to the user.

7. Features of the System

- Intelligent Query Processing

The AI agent can interpret natural language queries and provide meaningful responses.

- Real-Time Web Search

Integration with SerpAPI enables access to updated information from the internet.

- Contextual Memory

The system remembers previous interactions for better conversation flow.

- Automated Workflow

The entire process runs automatically through the n8n workflow.

- Scalable Design

Additional tools or APIs can be easily integrated.

8. Advantages

- Fast AI response generation

- Real-time information retrieval
- Easy workflow customization
- Visual workflow design
- Integration with multiple APIs

9. Limitations

- Dependent on external APIs
- Internet connection required
- Search API may have request limits
- AI responses depend on model accuracy

10. Future Enhancements

Future improvements may include:

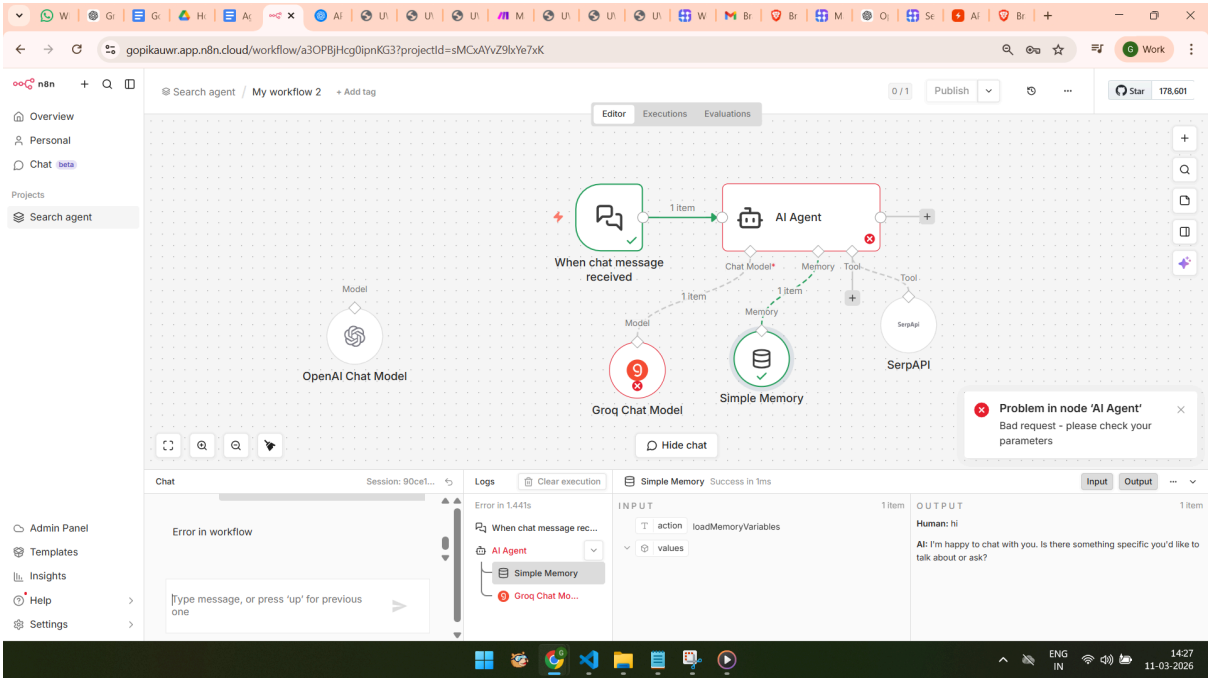
- Voice input and speech recognition
- Integration with messaging platforms (WhatsApp, Telegram)
- Multi-language support
- Database storage for conversation history
- Custom knowledge base integration
- Advanced reasoning AI models

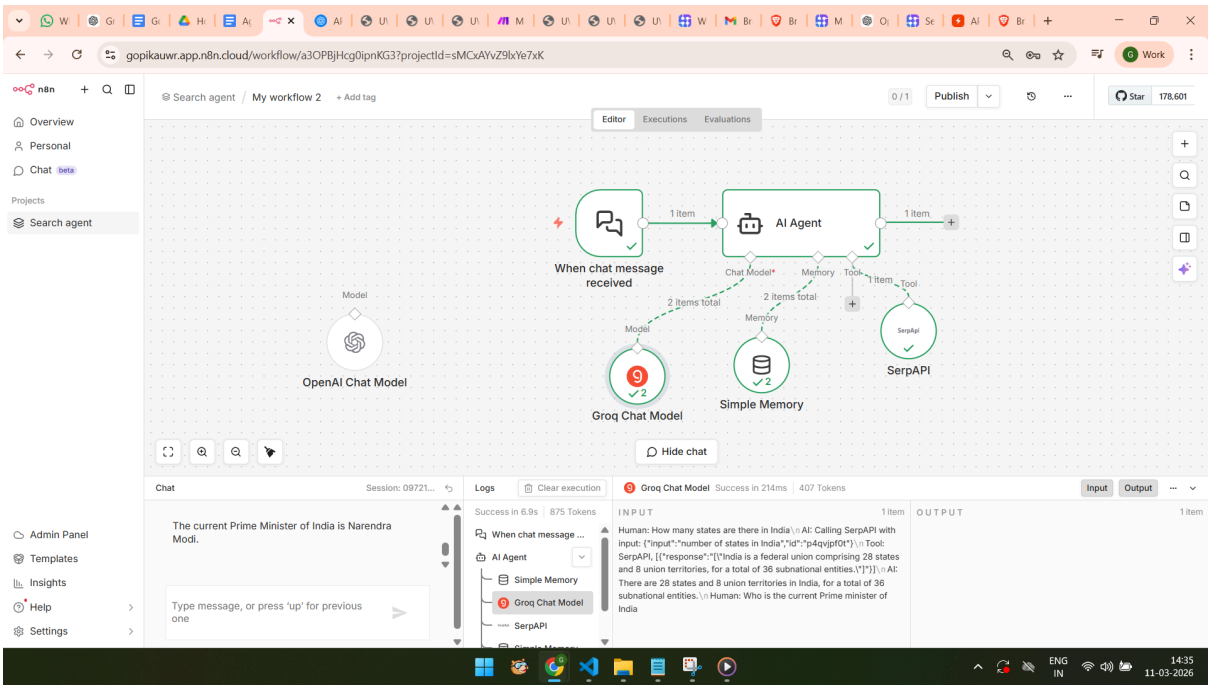
11. Conclusion

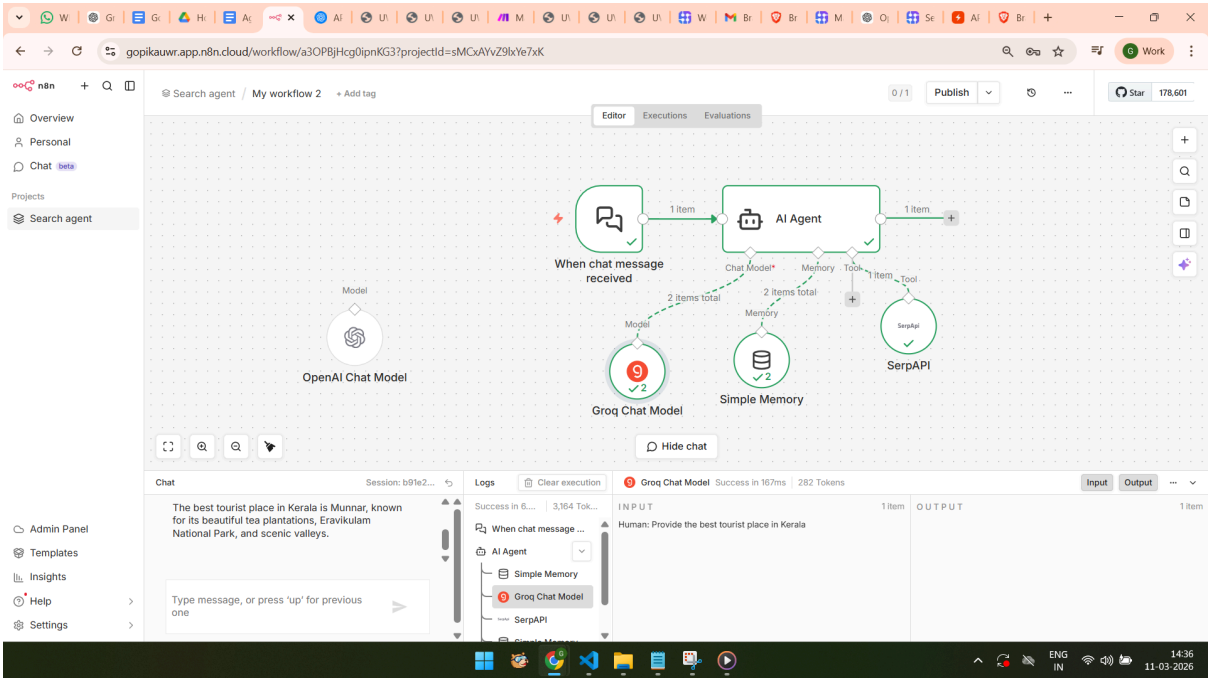
This project demonstrates the development of an AI-powered search assistant using a workflow automation platform. By integrating a language model, memory component, and web search API, the system provides intelligent responses to user queries.

The project highlights the potential of AI agents in automating information retrieval tasks and improving user interaction with digital systems.

12. Screenshots



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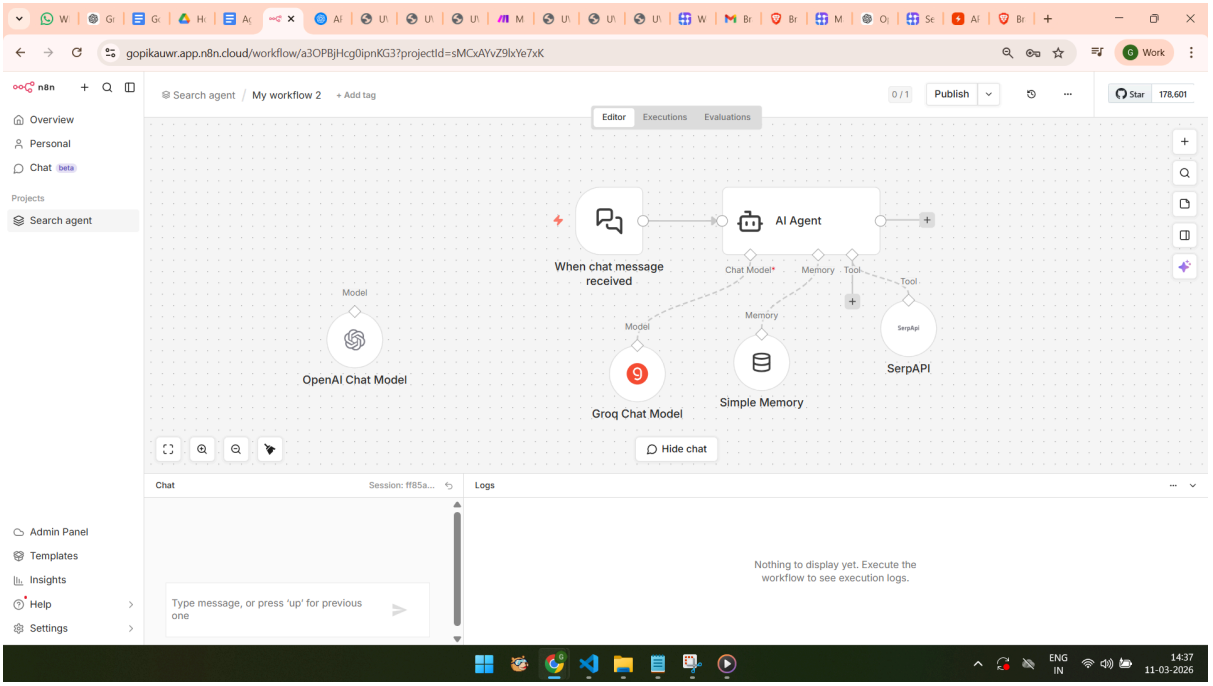
n8n
Search agent / My workflow 2
+ Add tag
Publish
Star 178,601
Overview
Personal
Chat beta
Projects
Search agent
Editor
Executions
Evaluations
When chat message received
1 item
AI Agent
1 item
Chat Model*
Memory
Tool
2 items total
2 items total
1 item
Tool
Model
OpenAI Chat Model
Model
Groq Chat Model
Memory
Simple Memory
SerpAPI
Hide chat
Chat
Session: b91e2...
The best tourist place in Kerala is Munnar, known for its beautiful tea plantations, Eravikulam National Park, and scenic valleys.
Type message, or press 'up' for previous one
Logs
Clear execution
Success in 6.... 3,164 Tok...
When chat message ...
AI Agent
Simple Memory
Groq Chat Model
SerpAPI
Groq Chat Model Success in 167ms 282 Tokens
Input
Output
INPUT
1 item
Human: Provide the best tourist place in Kerala
OUTPUT
1 item
Admin Panel
Templates
Insights
Help
Settings
ENG IN
14:36
11-03-2026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When chat message received
AI Agent
Chat Model*
Memory
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Tool
Model
OpenAI Chat Model
Model
Groq Chat Model
Memory
Simple Memory
SerpAPI
Hide chat
Chat
Session: ff85a...
Logs
Nothing to display yet. Execute the workflow to see execution logs.
Type message, or press 'up' for previous one
Admin Panel
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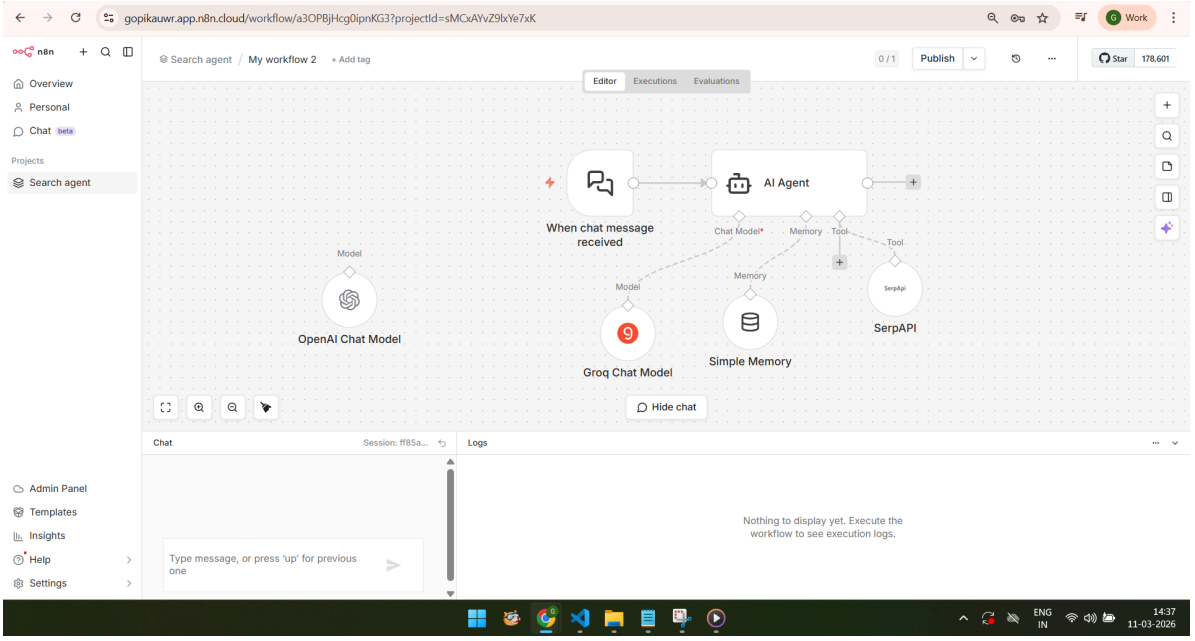
ENG IN
14:37
11-03-2026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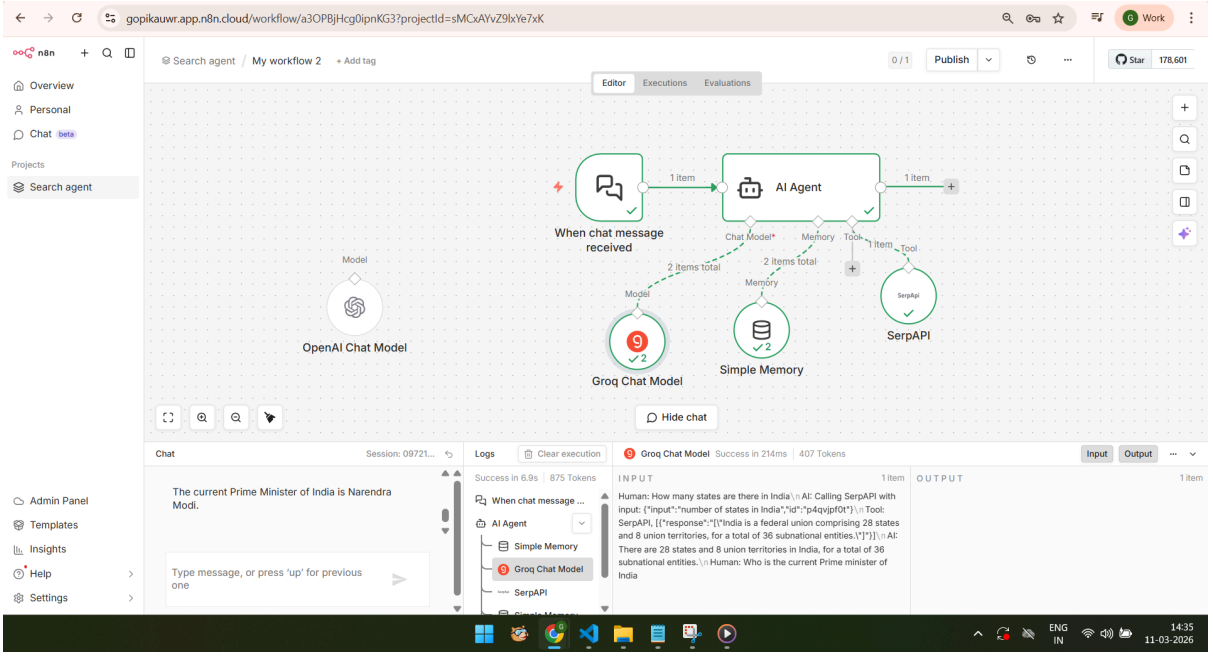
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1 item
AI Agent
Chat Model*
Memory
Tool
2 items total
Model
OpenAI Chat Model
Groq Chat Model
Simple Memory
SerpAPI
Hide chat
Chat
Session: 09721...
Logs
Clear execution
Groq Chat Model Success in 214ms 407 Tokens
Input
Output
Success in 6.9s 875 Tokens
When chat message ...
AI Agent
Simple Memory
Groq Chat Model
SerpAPI
INPUT
1 item
Human: How many states are there in India\n AI: Calling SerpAPI with input: {"input":"number of states in India","id":"p4qvjpf0t"}\n Tool: SerpAPI, [{"response":"[\"India is a federal union comprising 28 states and 8 union territories, for a total of 36 subnational entities.\"]"}]\n AI: There are 28 states and 8 union territories in India, for a total of 36 subnational entities.\n Human: Who is the current Prime minister of India
OUTPUT
1 item
The current Prime Minister of India is Narendra Modi.
Type message, or press 'up' for previous one
Admin Panel
Templates
Insights
Help
Settings
ENG IN
14:35
11-03-2026